

Exact Walsh Traces and Krawtchouk Compression for the Ehrenfest Hypercube Walk

Route-A structural certificate C171

Abstract

For every integer $d \geq 1$, we diagonalize the Ehrenfest coordinate-flip operator on the d -dimensional sign cube. Every Walsh character indexed by S has eigenvalue $1 - 2|S|/d$, with binomial multiplicity. This gives exact power traces, a factored determinant, and every vertex return probability; odd returns vanish by bipartiteness. Hamming weight is an exact Markov lumping to a $(d+1)$ -state reversible birth–death chain. Its symmetric Jacobi similarity has off-diagonal entries $\sqrt{(k+1)(d-k)}/d$, and Krawtchouk polynomials provide its simple spectrum, retaining every distinct eigenvalue of the original 2^d -state operator. The result is an all-parameter source theorem, not a finite extrapolation.

Keywords: Ehrenfest chain; hypercube dynamics; Walsh characters; Krawtchouk polynomials; reversible Markov chains; exact traces.

中文摘要

对任意整数 $d \geq 1$ ，本文精确对角化 d 维符号超立方体上的 Ehrenfest 坐标翻转算子。由集合 S 标记的 Walsh 特征函数具有特征值 $1 - 2|S|/d$ ，其重数为二项式系数。由此得到精确幂迹、因子化行列式与任意顶点的返回概率；二部结构使所有奇数时刻返回概率为零。汉明重量给出到 $d+1$ 状态可逆生灭链的精确马尔可夫压缩。其对称 Jacobi 相似矩阵的非对角元为 $\sqrt{(k+1)(d-k)}/d$ ，而 Krawtchouk 多项式给出单谱，从而保留原 2^d 状态算子的每个不同特征值。该结论是全参数源侧定理，不是有限样本外推。

关键词：Ehrenfest 链；超立方体动力学；Walsh 特征；Krawtchouk 多项式；可逆马尔可夫链；精确迹。

1 Frozen dynamics

Let $X_d = \{-1, +1\}^d$, let $x^{(i)}$ flip coordinate i , and set

$$(P_d f)(x) = \frac{1}{d} \sum_{i=1}^d f(x^{(i)}) \quad \text{on } L^2(X_d, 2^{-d} \#). \quad (1)$$

One coordinate flip is one Markov tick. For $S \subseteq \{1, \dots, d\}$, write $\chi_S(x) = \prod_{i \in S} x_i$.

Theorem 1 (complete Walsh certificate). For every $d \geq 1$, the Walsh characters form an orthonormal eigenbasis and

$$P_d \chi_S = \lambda_{|S|} \chi_S, \quad \lambda_j = 1 - \frac{2j}{d}, \quad \text{mult}(\lambda_j) = \binom{d}{j}. \quad (2)$$

Consequently, for every $n \geq 0$,

$$\text{Tr } P_d^n = \sum_{j=0}^d \binom{d}{j} \lambda_j^n, \quad (3)$$

$$\det(I - zP_d) = \prod_{j=0}^d (1 - z\lambda_j)^{\binom{d}{j}}. \quad (4)$$

Proof. A flip contributes $-\chi_S$ for the $|S|$ coordinates in S and $+\chi_S$ for the other $d - |S|$. Averaging proves (2). Walsh orthogonality and the count $\binom{d}{j}$ give a complete basis; (3)–(4) follow from the finite spectral theorem. $\square$

For $|z| < 1$, the scalar power series for $-\log(1 - z\lambda_j)$ gives the exact trace–log relation

$$-\log \det(I - zP_d) = \sum_{n \geq 1} \text{Tr}(P_d^n) \frac{z^n}{n}. \quad (5)$$

This is a weighted closed-Markov-walk identity. Because P_d is an average rather than a deterministic map, these traces are not fixed-point counts and (5) is not asserted to be an Artin–Mazur zeta.

2 Return law and exact compression

Cube translations commute with P_d and act transitively, so all diagonal entries of P_d^n agree. Their sum is the trace, hence

$$P_d^n(x, x) = 2^{-d} \text{Tr } P_d^n. \quad (6)$$

Every flip reverses Hamming parity, proving that (6) vanishes for odd n .

Let k count negative coordinates. The exact lumped kernel is

$$Q(k, k+1) = \frac{d-k}{d}, \quad Q(k, k-1) = \frac{k}{d}. \quad (7)$$

For $\pi_k = 2^{-d} \binom{d}{k}$, the binomial identity $\binom{d}{k}(d-k) = \binom{d}{k+1}(k+1)$ proves detailed balance. Therefore $D_\pi^{1/2} Q D_\pi^{-1/2}$ is symmetric with upper entry $\sqrt{(k+1)(d-k)}/d$.

Theorem 2 (Krawtchouk compression). Define

$$K_j(k) = \sum_r (-1)^r \binom{k}{r} \binom{d-k}{j-r}.$$

Then $QK_j = (1 - 2j/d)K_j$. Thus Q has the $d+1$ distinct eigenvalues in (2), each once, and compresses the 2^d -state operator without losing a distinct eigenvalue.

Proof. The generating function is $\sum_j K_j(k)t^j = (1-t)^k(1+t)^{d-k}$. Substitution of $k \pm 1$ using (7) equals the original series minus $(2t/d)$ times its derivative. Comparing t^j coefficients gives the eigenvalue equation. The eigenvalues are distinct, so the K_j form a basis. $\square$

3 Boundary

For $d = 1$, the theorem reduces to the deterministic two-cycle with spectrum $\{1, -1\}$. At even d , zero has multiplicity $\binom{d}{d/2}$; its factor in (4) is one and contributes no polynomial degree. At $n = 0$, (3) and (6) give 2^d and one, respectively. The unit eigenvalue makes $|z| < 1$ the natural domain stated for (5).

Exact ledgers through $d = 18, n = 24$, independent reconstruction, brute closed-walk enumeration for $d \leq 7, n \leq 8$, symbolic Krawtchouk checks, byte replay and repaired-hash mutations test the implementation; they do not prove the theorem. No target spectrum or arithmetic conclusion is asserted.